Transfer: energy and work

antocciphysics | TEACHER KEY | CPA

Learning target: Choose and justify energy methods in new physical situations.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 19; 3 course days later.*
Both become 1/9 of their original values; both scale with v^2.

02. Explain a worked example (worked)
Gravitational loss 40 J splits into kinetic 30 and thermal 10 J.

03. Look and explain (model)
Gravitational U decreases, K increases; height vertical.

04. Build the solution (practice)
40 J each.

05. Use the idea (reason)
Thermal, sound, etc; not destroyed.

06. Choose a method (mixed) *Review: Unit 4, Day 10; 12 course days later.*
K=20 J.

07. Exit check (exit)
20 W then 10 W; same energy, half power.

08. STRETCH: Apply it another way (stretch)
Reduce drop to 1.5 m with 10 J loss, or keep 2 m drop with 20 J loss; other valid accounts accepted.

Teaching note: Self-contained daily transfer worksheet. Keep any separately assigned formal assessment independent.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 19; 3 course days later.*
Coupled energy becomes 3/5≈0.6 of the original; multiply the mass and fraction factors.

02. Explain a worked example (worked)
Gravitational loss 40 J splits into kinetic 30 and thermal 10 J.

03. Represent and interpret (model)
Gravitational U decreases, K increases; height vertical.

04. Solve and justify (practice)
Initial 60 J; bottom 16 J; loss 44 J;$\frac{1}{2}kx^2=16\rightarrow x=.40$ m.

05. Analyze the reasoning (reason)
Thermal, sound, etc; not destroyed.

06. Choose a method (mixed) *Review: Unit 4, Day 10; 12 course days later.*
$h=v_i^2/(2\ g)=20$ m.

07. Exit check (exit)
10 m/s; no atmosphere, ideal central gravity, no further thrust, ignores rotation/other bodies.

08. STRETCH: Apply it another way (stretch)
Initial K 6 plus gravitational 30 minus loss 11=25 J; $\frac{1}{2}kx^2=25$ gives$x=0.5$ m. At maximum compression speed zero.

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